

DS-5343

Human Computer Interaction and Evaluation

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UNIDAD 3



Outlook



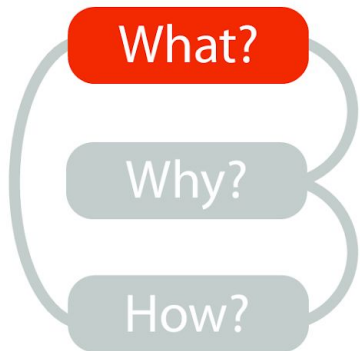
1. Human Computer Interaction
2. Techniques for evaluation
3. Usability studies



Session Objectives

Understand the fundamentals of Human-Computer Interaction (HCI) and explore key evaluation techniques used in the design and analysis of interactive systems. Experiments, interviews, surveys, case studies, and usability studies to assess user experience and system effectiveness.





What?

Datasets

➔ Data Types

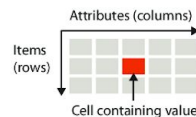
➔ Items ➔ Attributes ➔ Links ➔ Positions ➔ Grids

➔ Data and Dataset Types

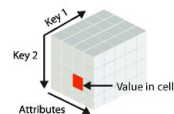
Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

➔ Dataset Types

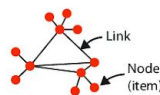
➔ Tables



➔ Multidimensional Table



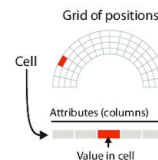
➔ Networks



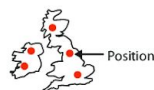
➔ Trees



➔ Fields (Continuous)



➔ Geometry (Spatial)



Attributes

➔ Attribute Types

➔ Categorical



➔ Ordered

➔ Ordinal



➔ Quantitative

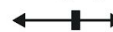


➔ Ordering Direction

➔ Sequential



➔ Diverging



➔ Cyclic



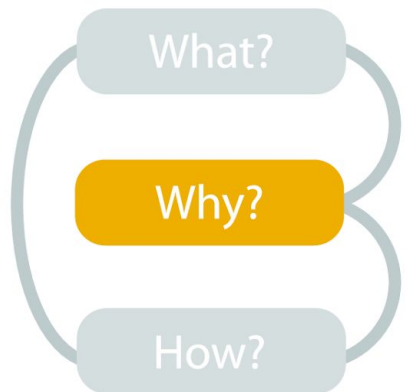
➔ Dataset Availability

➔ Static

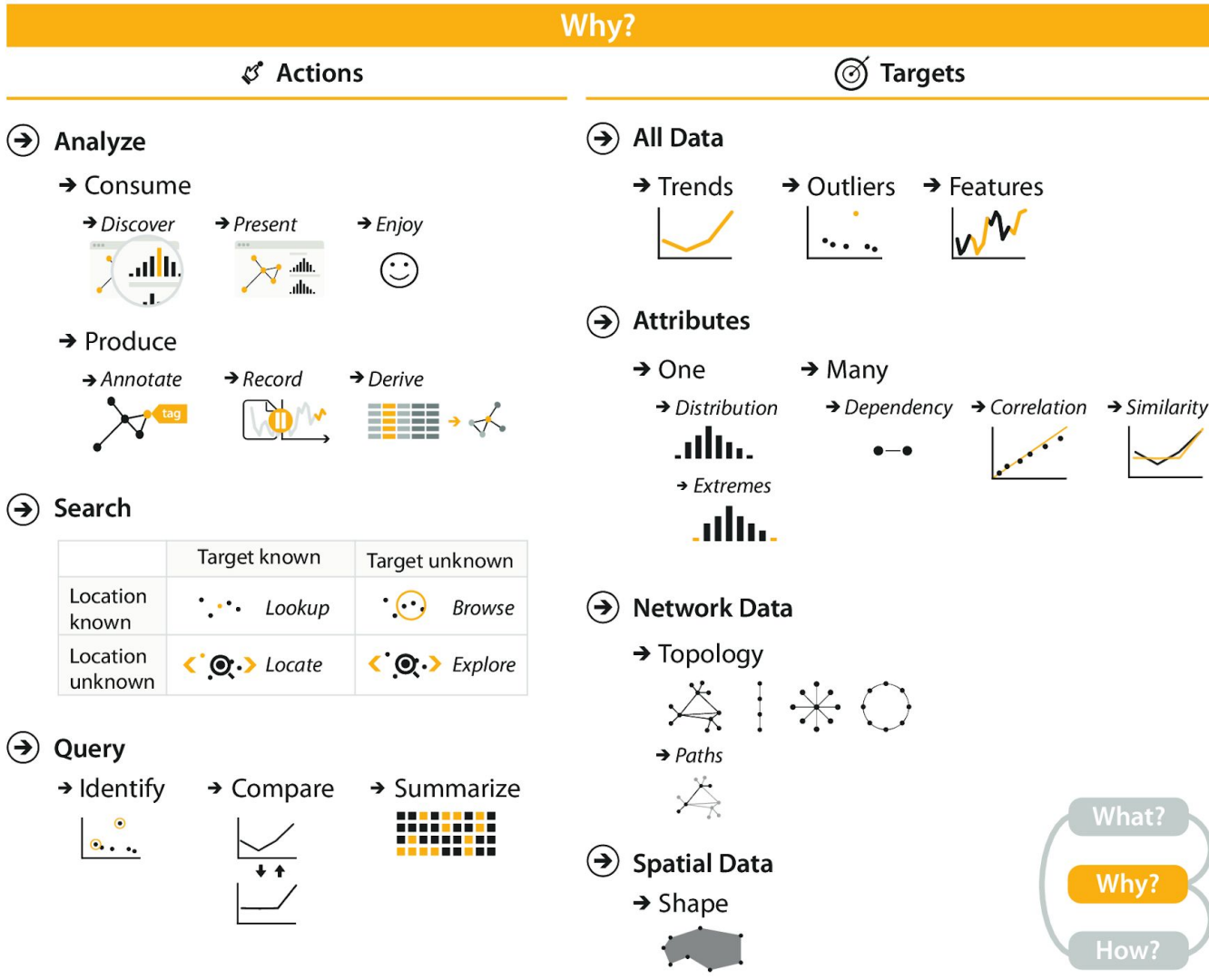


➔ Dynamic





- {action, target} pairs
 - *discover distribution*
 - *compare trends*
 - *locate outliers*
 - *browse topology*



How?

Encode

➔ Arrange

➔ Express



➔ Separate



➔ Order



➔ Align



➔ Use



➔ Map

from **categorical** and **ordered** attributes

➔ Color

➔ Hue



➔ Saturation



➔ Luminance



➔ Size, Angle, Curvature, ...



➔ Shape



➔ Motion

Direction, Rate, Frequency, ...



Manipulate

➔ Change



➔ Select

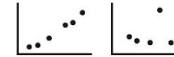


➔ Navigate



Facet

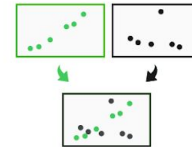
➔ Juxtapose



➔ Partition



➔ Superimpose



Reduce

➔ Filter



➔ Aggregate



➔ Embed



What?

Why?

How?

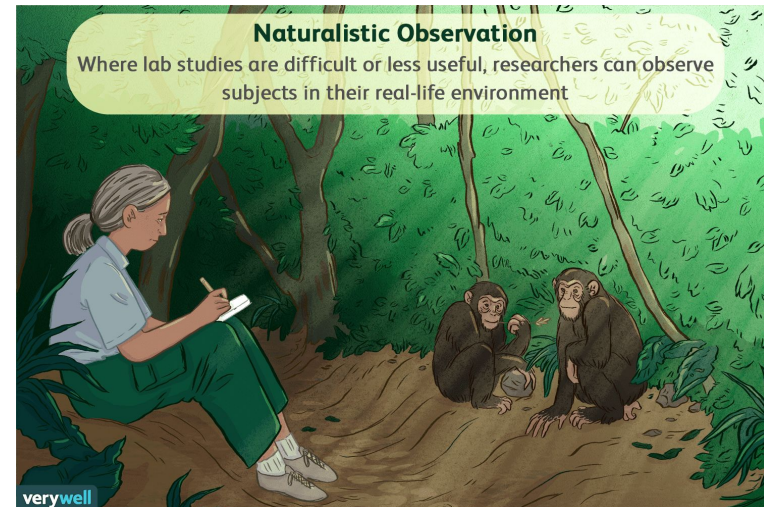
1.

Evaluation



Evaluation

- Controlled experiments (Papers)
- Natural settings (User Environments)
- Any setting not involving users (expert reviews)



Expert Reviews

- Design experts
- Visualization experts
- Usability experts
- Domain Experts

Types of Expert Reviews

- Heuristic evaluation (golden rules)
- Guidelines review
- Consistency inspection
- Cognitive walkthrough
- Metaphors of human thinking
- Formal usability inspection (courtroom style)
- Accessibility inspection

Eight Golden Rules of Design

1. Strive for consistency: Keep the interface predictable and uniform.
2. Cater for universal usability: Make the system usable for all types of users
3. Offer informative feedback: Give clear and immediate responses to user actions
4. Design dialogs to yield closure: Provide a clear beginning, process, and completion.
5. Prevent errors: Avoid mistakes before they happen.
6. Permit easy reversal of actions: Allow users to undo actions effortlessly
7. Support internal locus of control: Let users feel in control of the system
8. Reduce short-term memory load: Don't force users to remember too much at once



2.

Controlled Experiments



Controlled Experiments

- Experiments in the lab
- Controlled confounding variables
- Measure one or more quantitative variables
 - Usability testing
 - Living labs

What to Measure?

What to Measure?

- Time to learn
- Speed of performance
- Rate of errors
- Retention over time
- Subject satisfaction

Variables

- Independent variables (causes)
- Dependent variables (effect)
- Extraneous variables (that can affect the experiment)

Effects of distribution on Effectiveness of dataviz

In addition to the choice of visual encodings, the effectiveness of a data visualization may vary with the *analytical task* being performed and the *distribution of data values*.)

Are marks and channels the most important factor on effectiveness?

Experiment

To better assess these effects and create refined rankings of visual encodings, we conduct an experiment measuring subject performance across task types (e.g., comparing individual versus aggregate values) and data distributions (e.g., with varied cardinalities and entropies).

Measure performance ⊗ different tasks ⊗ data cardinality ⊗ data entropy



Tasks and conditions

Cardinality

3 20

Read Value

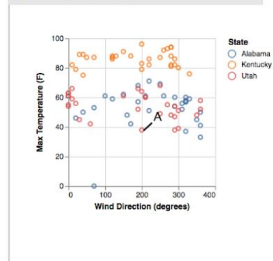
"What is the Max Temperature of the data point A?"

Task

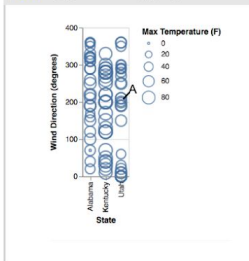
Compare Averages

"Which State has greater average Max Temperature?"

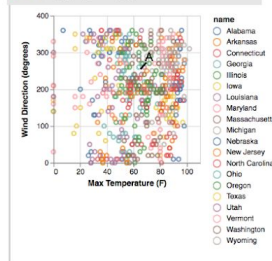
$Q_1:y, Q_2:x, N:color$



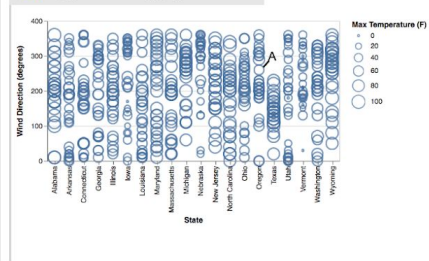
$Q_1:size Q_2:y, N:x$



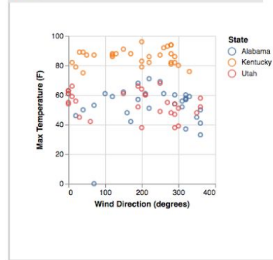
$Q_1:y, Q_2:x, N:color$



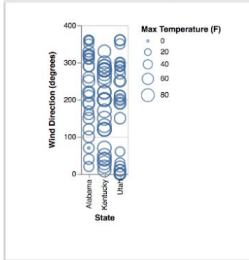
$Q_1:size Q_2:y, N:x$



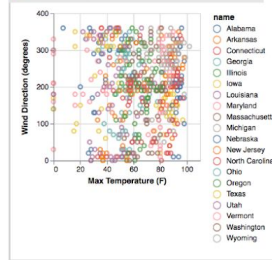
$Q_1:y, Q_2:x, N:color$



$Q_1:size Q_2:y, N:x$



$Q_1:y, Q_2:x, N:color$



$Q_1:size Q_2:y, N:x$

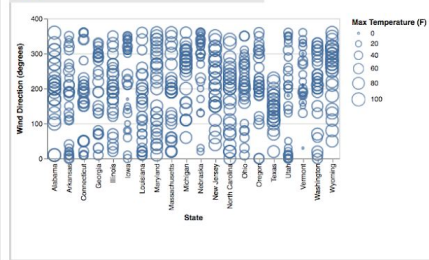
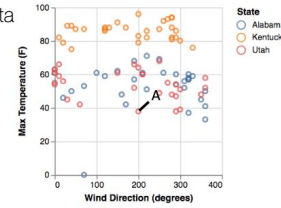


Figure 1: A subset of experiment stimuli across tasks (Read Value and Compare Averages) and cardinalities (3, 20). Each stimulus is labeled with its mapping from data fields (Q_1, Q_2, N) to visual channels ($x, y, color, size$).

Tasks

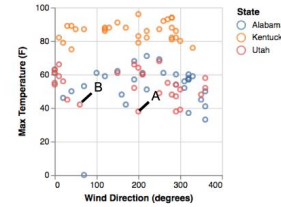
Read Value

What is the *Max Temperature* of the data point A?



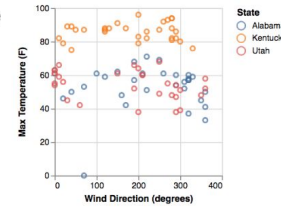
Compare Values

Which data point has **less** *Max Temperature*?



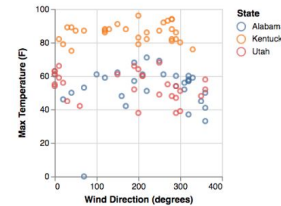
Find Maximum

Which State has the data point with the highest *Max Temperature*?



Compare Averages

Considering all data points for the State, which of the following two State has greater **average** *Max Temperature*?



Results

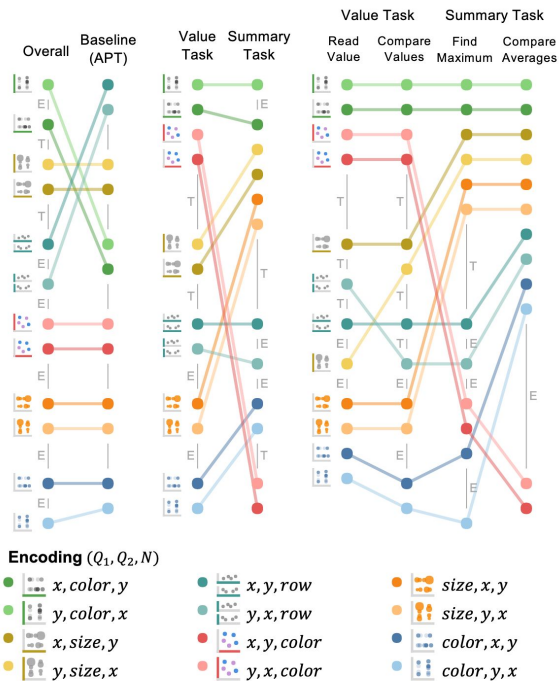


Figure 5: Visual encoding effectiveness rankings. Higher is better; groupings indicate encodings of identical rank. The left side compares value tasks and summary tasks with a baseline ranking by APT [Mac86]. The right side compares performance for all tasks, averaged across data distribution conditions. Gray lines indicate sig. differences ($p < 0.05$) between groups in terms of error (E) or completion time (T).

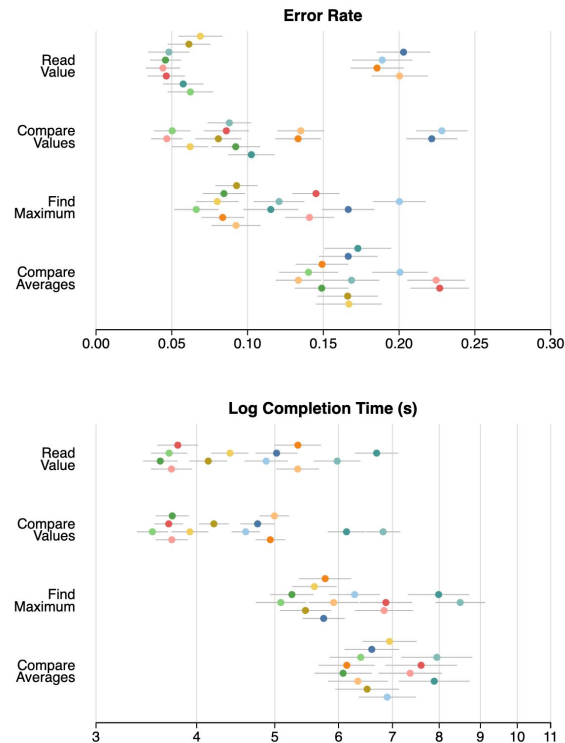


Figure 6: Bootstrapped means and 95% confidence intervals for error rates and log completion times across tasks. For example, $x, color, y$ and $y, color, x$ exhibit reasonable error rates but longer completion times.

Effects of tasks

- Position (x, y) conveys the primary quantities well.
- Size encoding performs well for summary tasks.
- Color encoding performs well for Compare Averages.
- Size & color exhibit asymmetric effects for Q1 vs. Q2.
- Faceted charts exhibit asymmetric performance.

3.

Usability Testing



Usability Testing



Natural Settings Involving Users

- Observation
- Interviews
- Logging

Triangulation

Different researchers observe the same effect.

Interviews

- Unstructured
- Structured
- Semi-structured
- Focus group
- Telephone/online interviews

Questionnaire

Like interviews but without the researcher present

3.

Likert Scales



Likert Scale

What do you think?

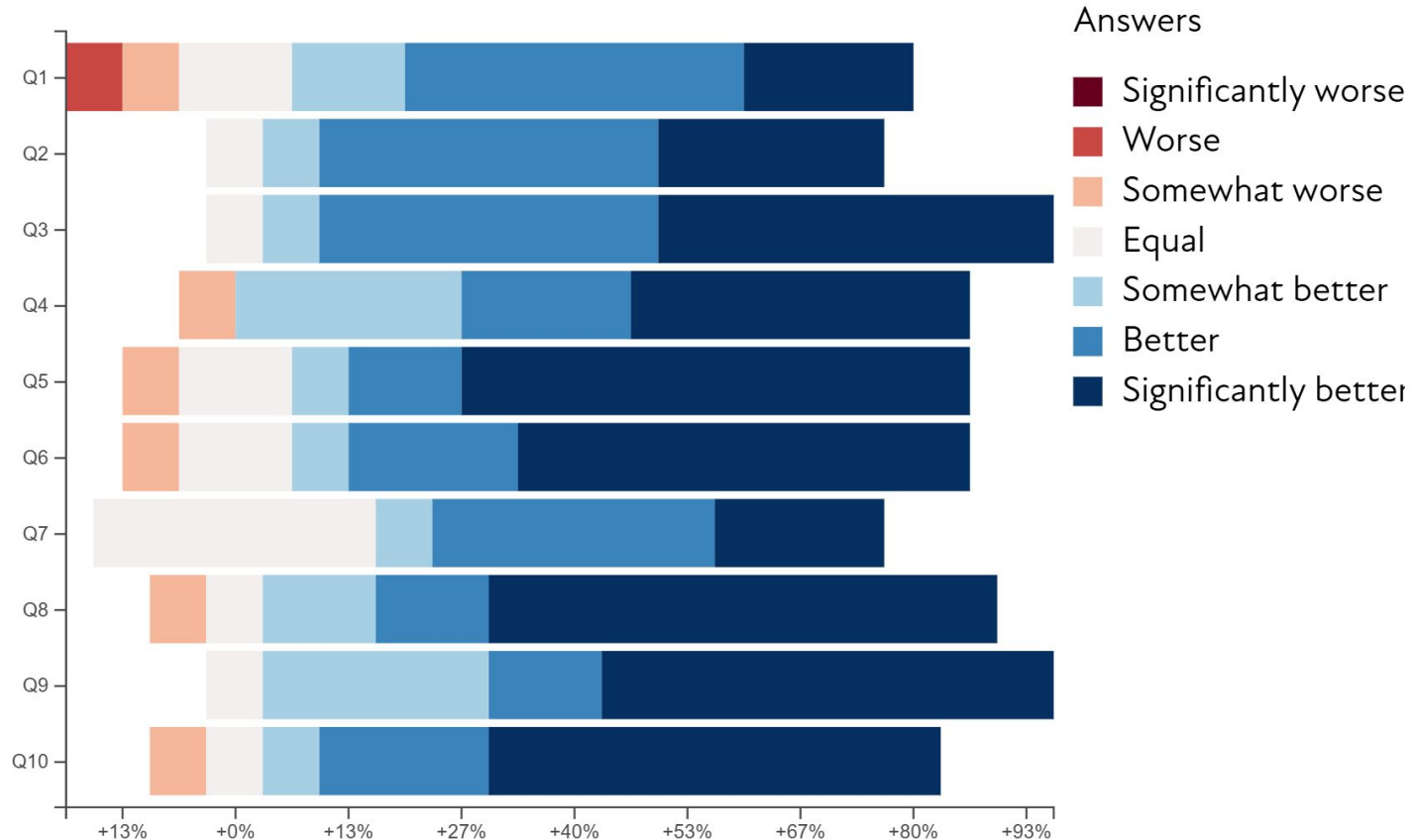
- Strongly disagree
- Disagree
- Okay
- Agree
- Strongly agree

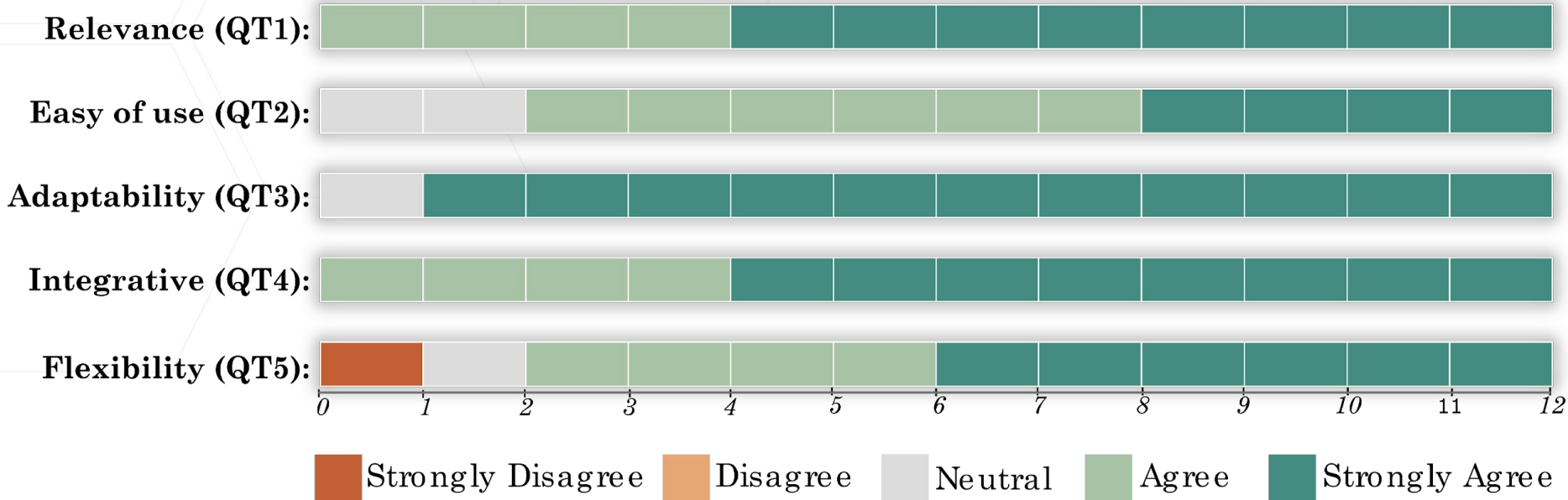
More About Likert Scales

What do you think?

- Can be 3, 5, 7, or more responses
- Continuous or discrete
- Middle response is the balance

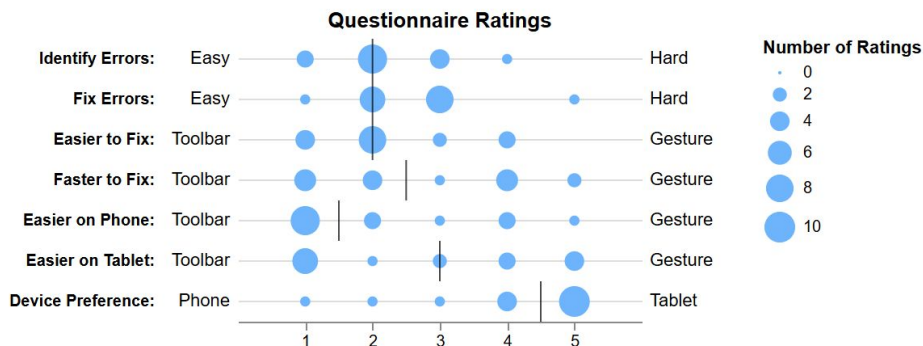
Likert Scales d3





Distributions and Medians of Likert Scale Ratings

Distributions and Medians of Likert Scale Ratings. (Figure 9 from [@jhoffswell](#) and [@zcliu](#)'s 'Interactive Repair of Tables Extracted from PDF Documents on Mobile Devices' – <http://idl.cs.washington.edu/files/2019-InteractiveTableRepair-CHI.pdf>).

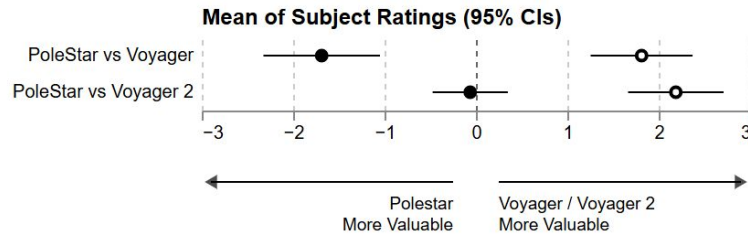


[View this example in the online editor](#)



Comparative Likert Scale Ratings

Comparing Likert scale ratings between two conditions. (Figure 10. from @kanitw et al.'s "Voyager 2: Augmenting Visual Analysis with Partial View Specifications" – <http://idl.cs.washington.edu/files/2017-Voyager2-CHI.pdf>).



4.

Other Methods



Observation

- User's setting
- Can be direct or indirect

Direct Observation in the Field

- Ethnography

Direct Observation in Controlled Environments

Think aloud techniques

Direct Observation: Tracking Users

Diaries

Interaction logs and web analytics

- Multi-dimensional
- In-depth
- Long-term
- Case studies

- Thirteen different case studies with nine agencies
- TreeVersity 2: UMD Budget 2010 - 2012

Focus groups

One researcher, many attendees

Prototyping

- Low vs. high fidelity?
- Read data
- Build scenarios, tell a story

Quatitative evaluation

Analysis of Variance (ANOVA) for comparing the means

5.

Running a Usability Study



Validity Checks

Earlier stages:

- Observe and interview target users (needs assessment)
- Design data abstraction/operation (data types, transformation, operations)
- Justify encoding/interaction design (design heuristics, perception research)
- Informal analysis/qualitative analysis of prototypes (task-based)
- Algorithm complexity analysis/evaluation

Validity Checks

Mid- and later stages:

- Qualitative analysis of system (task-based)
- Algorithm performance analysis
- Lab or crowdsourced user study
- Field study of the deployed system

Formal Usability Study

**GOAL: DOES THE VISUALIZATION ALLOW
THE USER/ANALYST TO PERFORM KEY
TASKS?**

Task-Oriented Visual Insights

- Basic insights:
 - Read a value
 - Identify extrema
 - Characterize distribution
 - Describe correlation
- Comparative insights:
 - Compare values
 - Compare extrema
 - Compare distribution
 - Compare correlation

Usability Study: Logistics

You will need:

- Visualization with test data loaded
- Consent form (if required)
- Task list
- Protocol (study procedures and debrief questions)
- Surveys/interviews and any additional data-collection instruments
- Audio or video recorder, notepad

How Many People Do You Need?

**"Lab" Doesn't Need to Mean a Formal
Lab**

Software for Collecting Audio/Video

- Video of user
- Screenshot of user actions
- Audio of entire session

Online Tools

- Surveys
- Mouse tracking/navigation tracking

6.

Prioritization



You've Collected Data

- Task completion
- Time on task
- Notes
- Interview responses
- Survey responses
- ...Then what?

What is the Analyst's Information Scent?

A term originating from Peter Pirolli and Stuart Card PARC looking at meaning Accounts for not just completion but also situating **WHAT** the person was doing, may wish to do think aloud as well. It's not always obvious what leads to the challenge.

Like a detective, not always obvious. Transacting to design.

MoSCoW Prioritization

- Must
- Should
- Could
- Won't

Conclusiones

Visualization techniques enable users to explore what features the model is focusing on and why. These tools help debug models, detect biases, and build trust, especially in sensitive domains like healthcare or safety.

They also facilitate communication with **non-experts** by translating complex behaviors into intuitive visuals.

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